CAN THO UNIVERSITY
COLLEGE OF INFORMATION AND COMMUNICATION TECHNOLOGY
OPERATING SYSTEMS (CT104H)
LAB #3

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- **Submission:** Students submit 1 file named *StudentName_ID_CT104H_Lab03.pdf* to the Google classroom (where *StudentName* is the student's name, and *ID* is the student's ID).
- **Instructions on how to present in the report file**

 For each question, students MUST provide the commands/scripts AND screenshots of the commands used and/or the content of files/scripts, CLEARLY. Note*: the screenshot needs to include the name of the Ubuntu Virtual machine.*

 Student creates an Ubuntu Virtual machine named UbuntuID (ID is the student's ID). For example, we have a virtual machine named Ubuntu20043, such that the student *ID* is 20043.

Question 0: Navigate to your home directory

Answer: $cd

Study from the files: Linux Shell Scripting Tutorial, and Shell Script (Chapter 2, 3), Advanced Bash-Scripting Guide (Chapter 13)

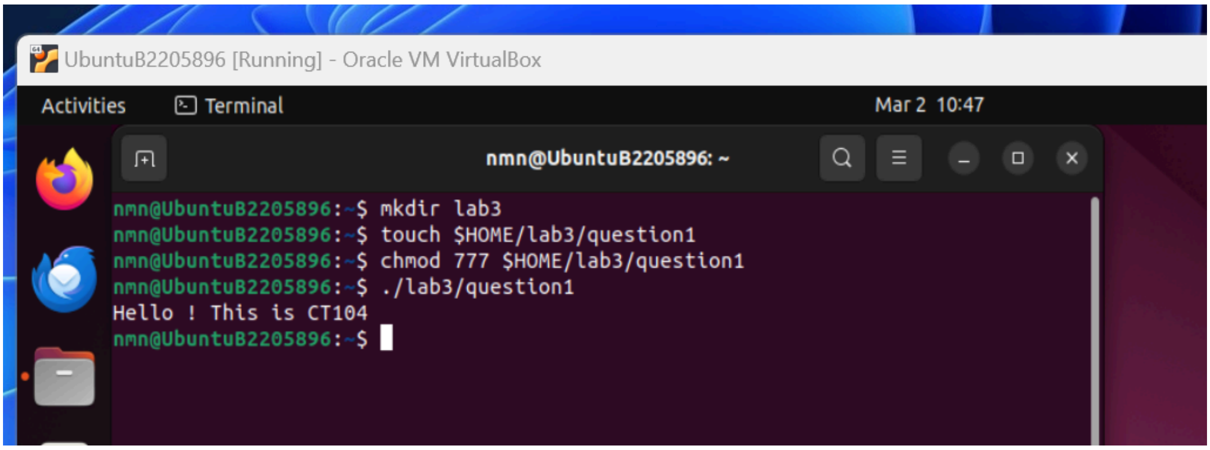
Question 1: Write a shell script that prints the message "Hello ! This is CT104" to the screen.

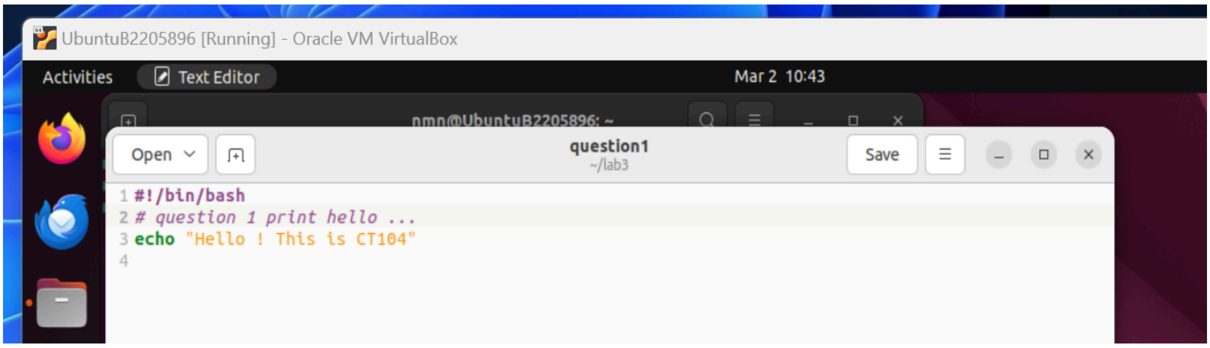
Answer:

#!/bin/bash

question 1 print hello

echo "Hello! This is CT104"





Question 2: Modify the shell script from Question 1 to include a variable. The variable will hold the content of the message.

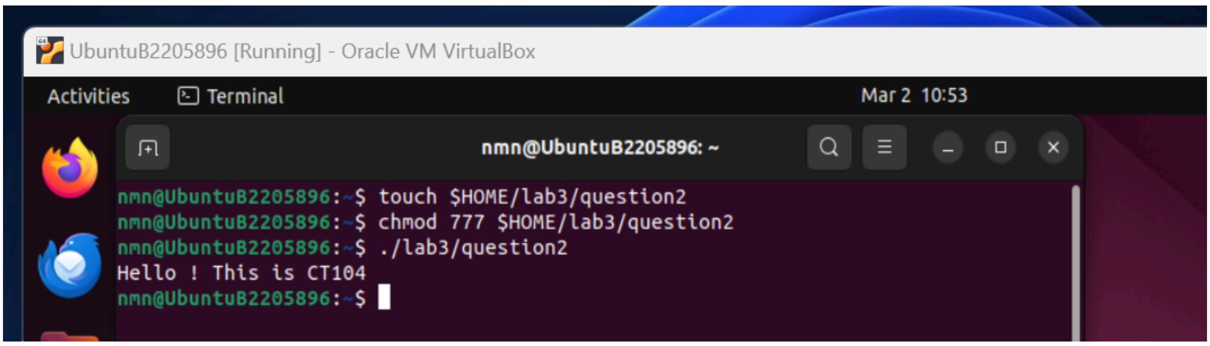
Answer:

#!/bin/bash

question 1 print hello … with variable

mess="Hello! This is CT104"

echo $mess



Question 3: Write a shell script that prints the following information to the screen

y is <*current date*>

<*username*>

current working directory is <*current directory*>

home directory is <*home directory*>

e press any key to finish <*wating to receive a key from user*>

********* Thank you

Answer:

#!/bin/bash

question 3 print

echo “Today is $date”

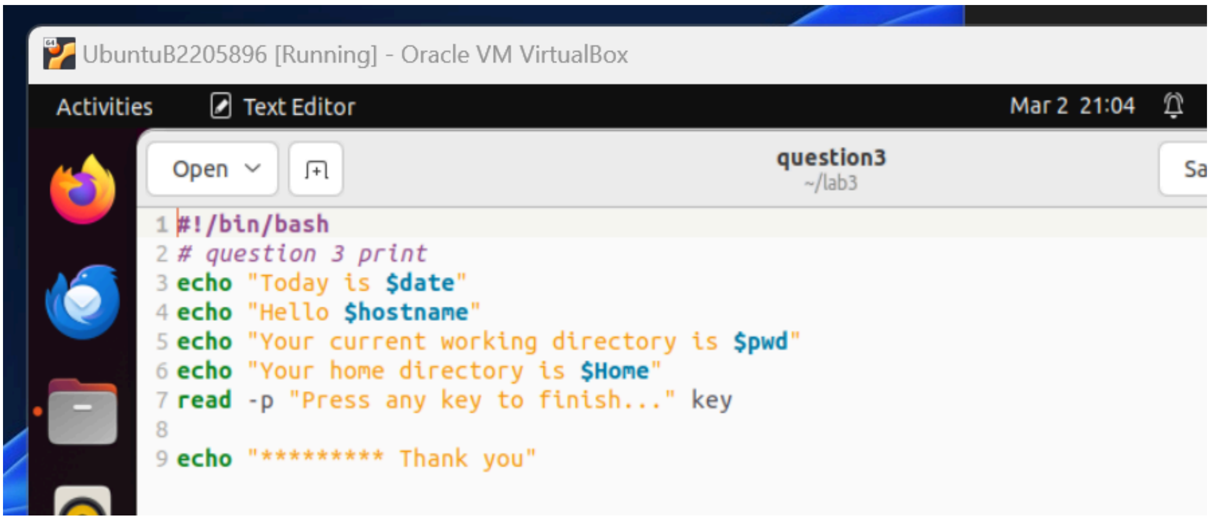
echo “Hello $hostname

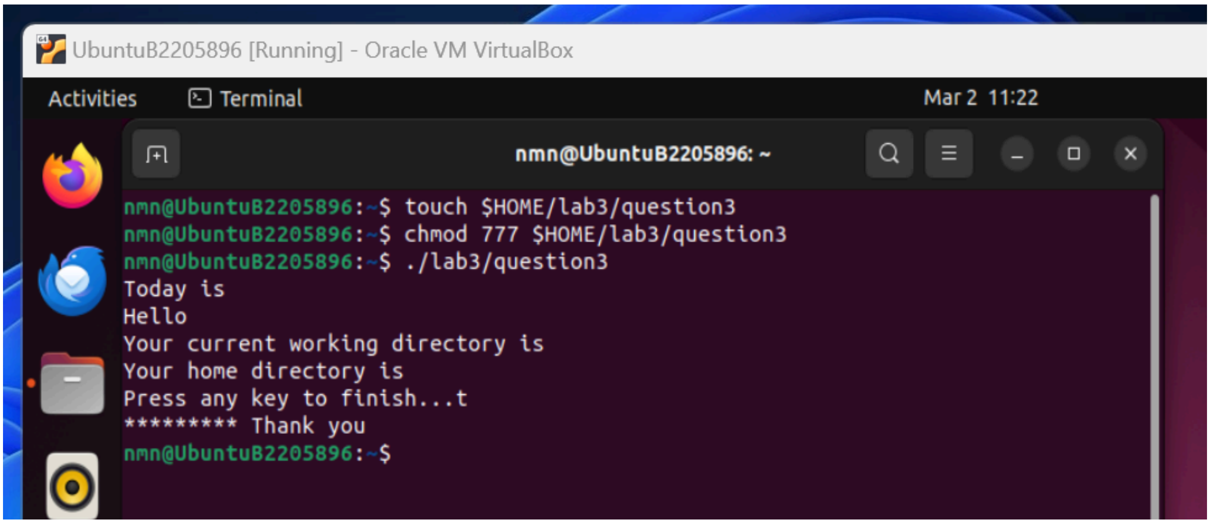
echo “Your current working directory is $pwd”

echo “Your home directory is $Home”

read -p “Press any key to finish…” key

echo “******** Thank you”





Question 4: Write a shell script that receives 2 numbers from a user, and calculates the following values

e input the first number: x = *x*

e input the second number: y = *y*

= *<x-y>*

) = *<x+y>*

) = *<x*y>*

= *<x/y>*

) = *<x%y>*

2

x!

Answer:

#!/bin/bash

question 4

get 2 number

read -p "Please input the first number: "x

read -p "Please input the second number: y

tinh gia thua

factorial=1

for (i=2; i<=x; i++));

do

factorial=$((factorial*i))

done

echo "(x-y)=$((x-y))"

echo "(x+y)=$((x+y))"

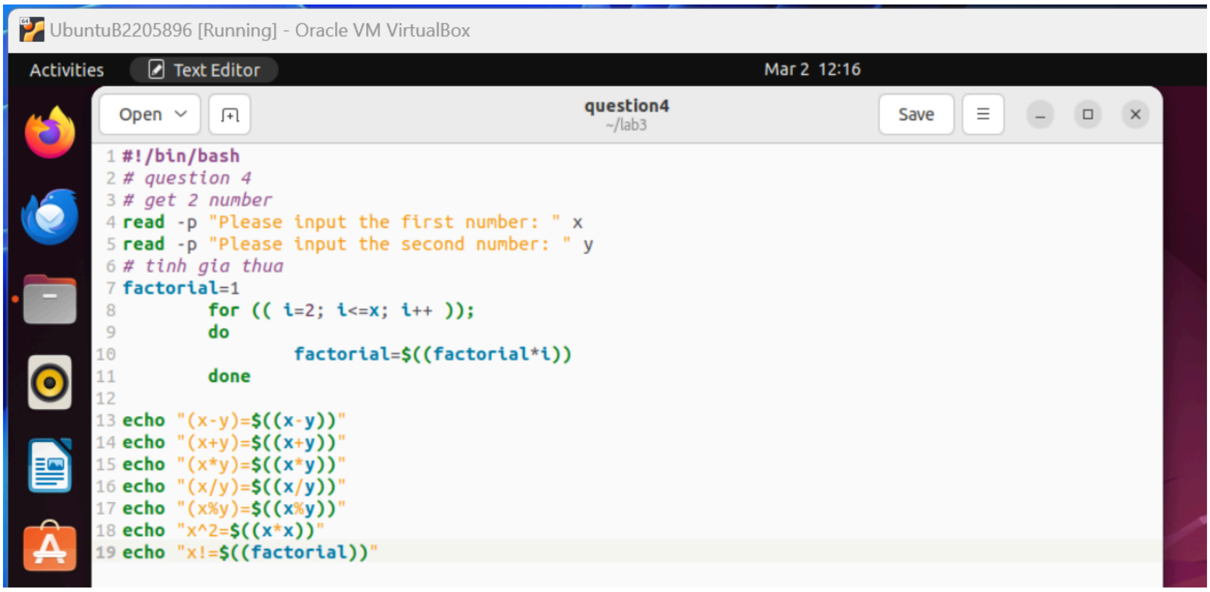
echo "(x*y)=$((x*y))"

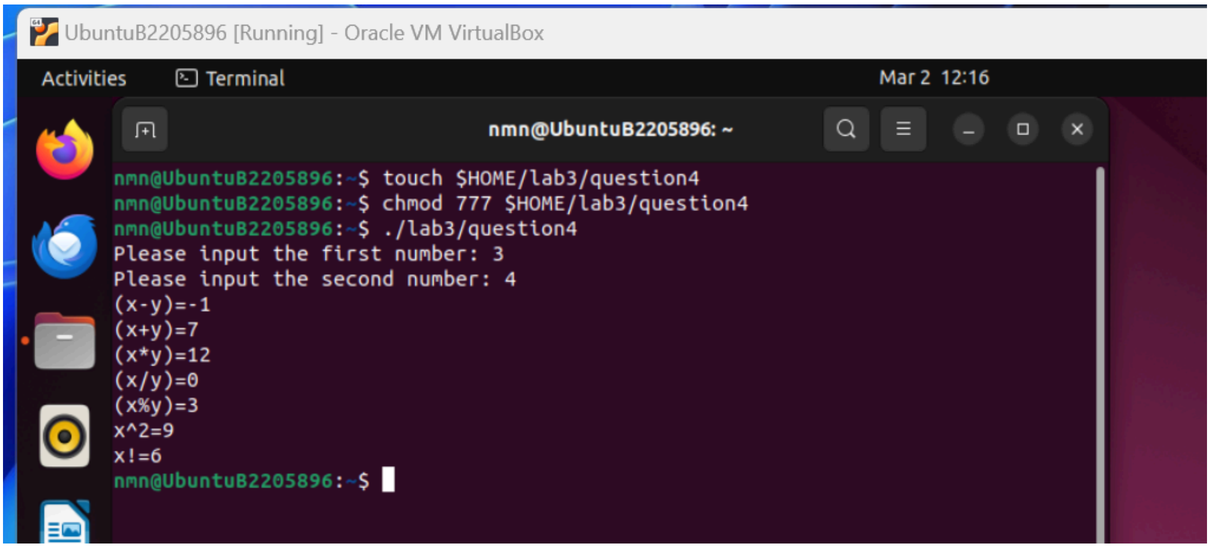
echo "(x/y)=$((x/y))"

echo "(xy)=$((x%y))"

echo "x^2=$((x*x))"

echo "x!=$((factorial))"





Question 5: Write a shell script that allows a user to input a name of a directory, and creates this directory for the user. If the directory is created, please print the message "The *name_of_directory* is created successfully" to the screen; otherwise, print the message "Can not create the *name_of_directory*" to the screen.

Answer:

#!/bin/bash

question 5

get directory_name

read -p "Please input name of directory: " direc_name

create the directory

mkdir $direc_name

check

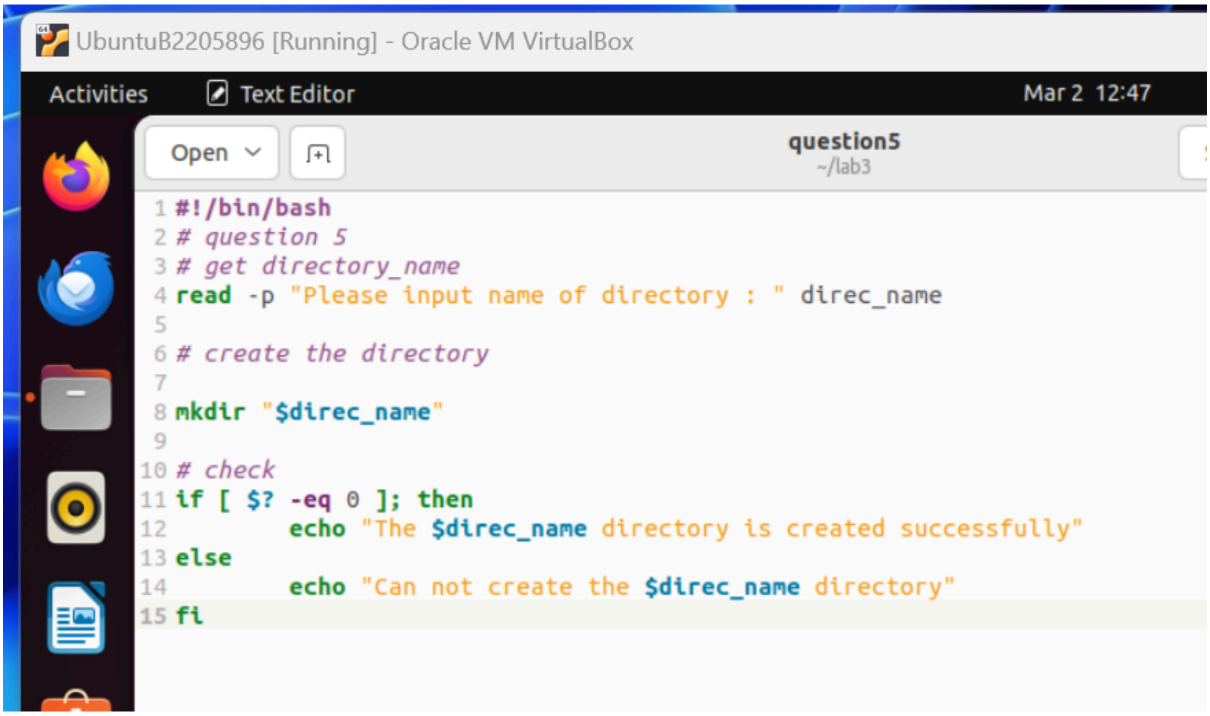
if [$? -eq 0]; then

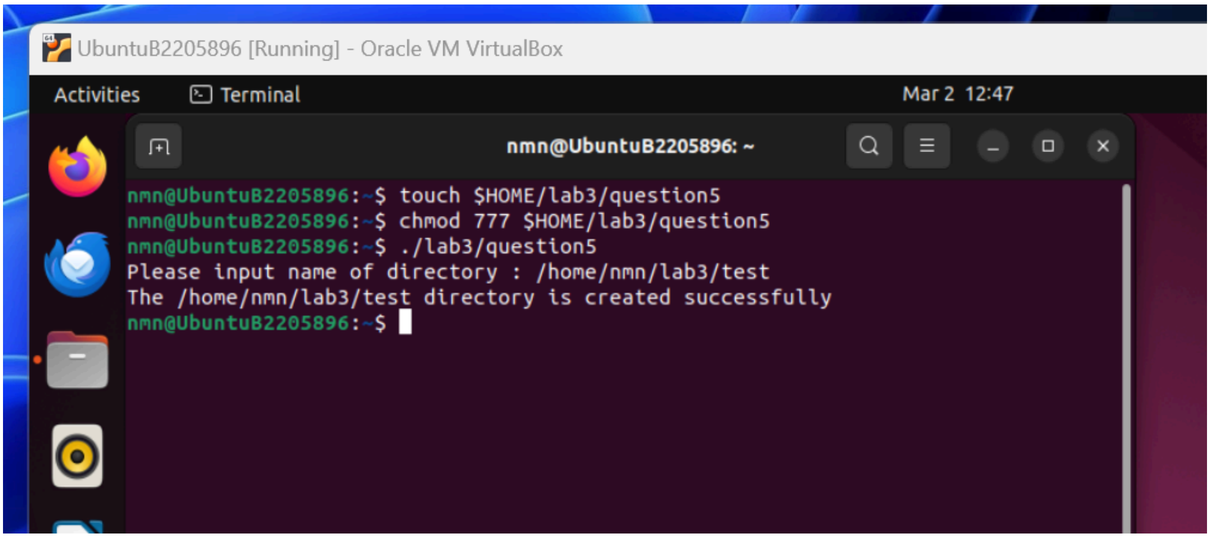
echo "The $direc_name directory is created successfully"

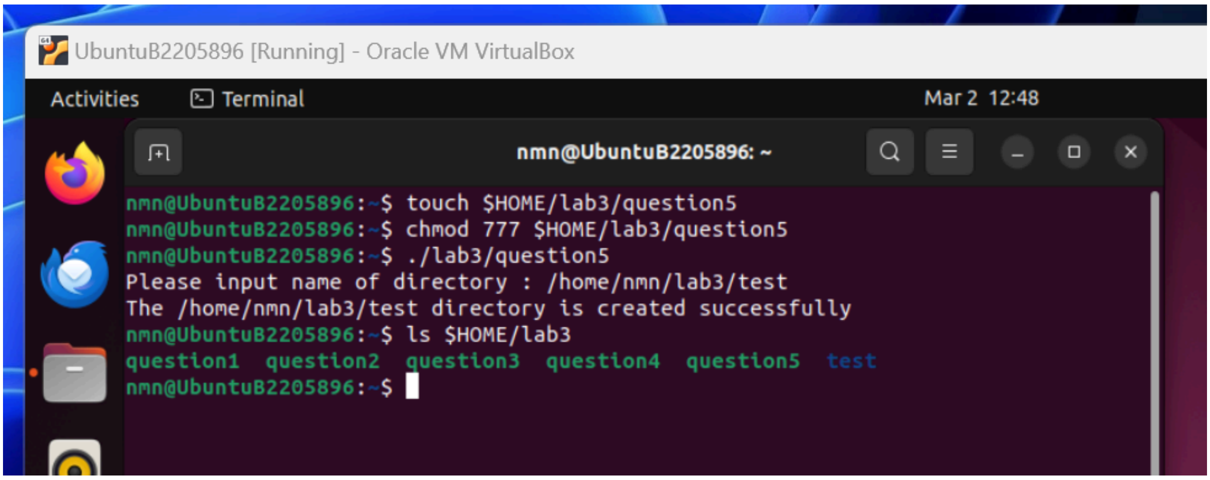
else

echo "Can not create the $direc_name directory"

fi







Question 6: Write a script to print the following symbols to the screen

```
*
**
***
****
*****
******
*******
********
*********
**********
```

1	2	3	4	5
1	2	3	4	5
1	2	3	4	5
1	2	3	4	5
1	2	3	4	5

Answer:

```
#!/bin/bash
# question 6 print star and number3
 # print star
for((i=1; i<= 10; i++ ))
do
for(( j=1; i<=i; j++ ))
do
echo -n "*"
```

done

echo “”

done

for((i=1; i<=5; i++))

do

for((j=1; i<=5; j++))

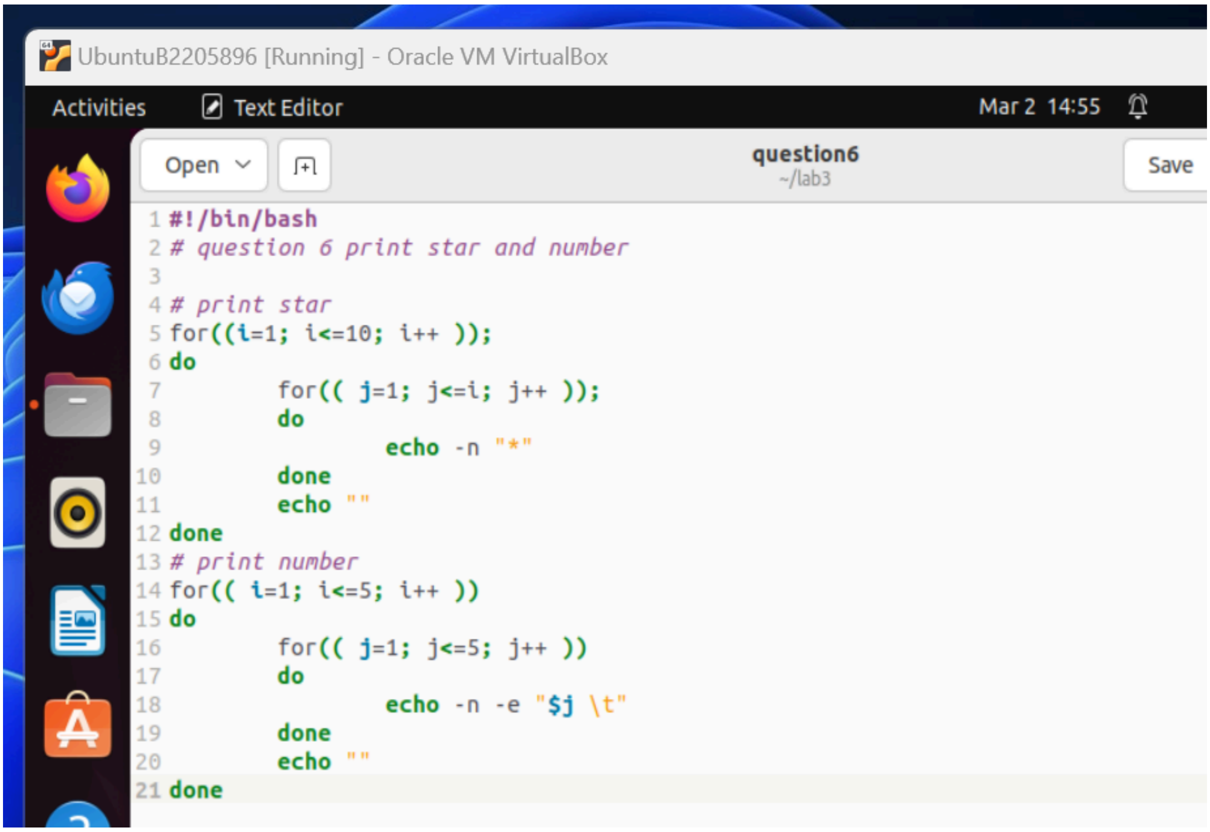
do

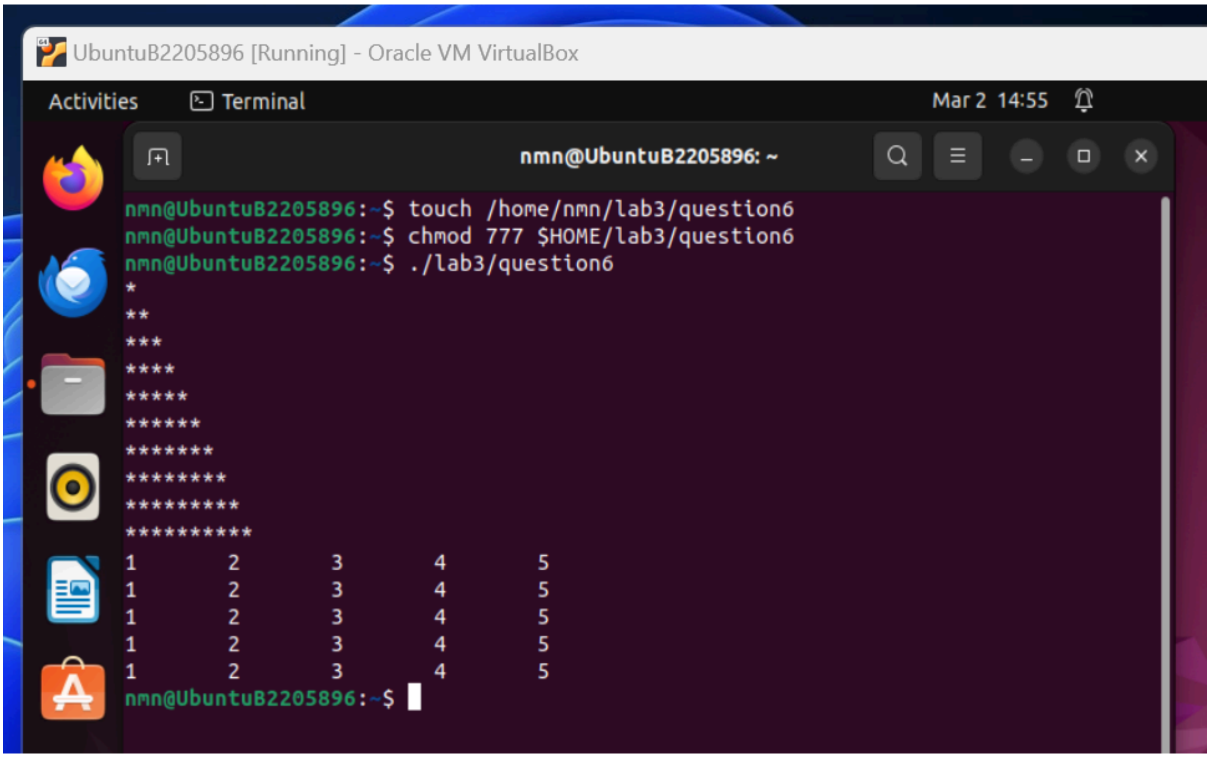
echo -n -e "$j \t"

done

echo “”

done

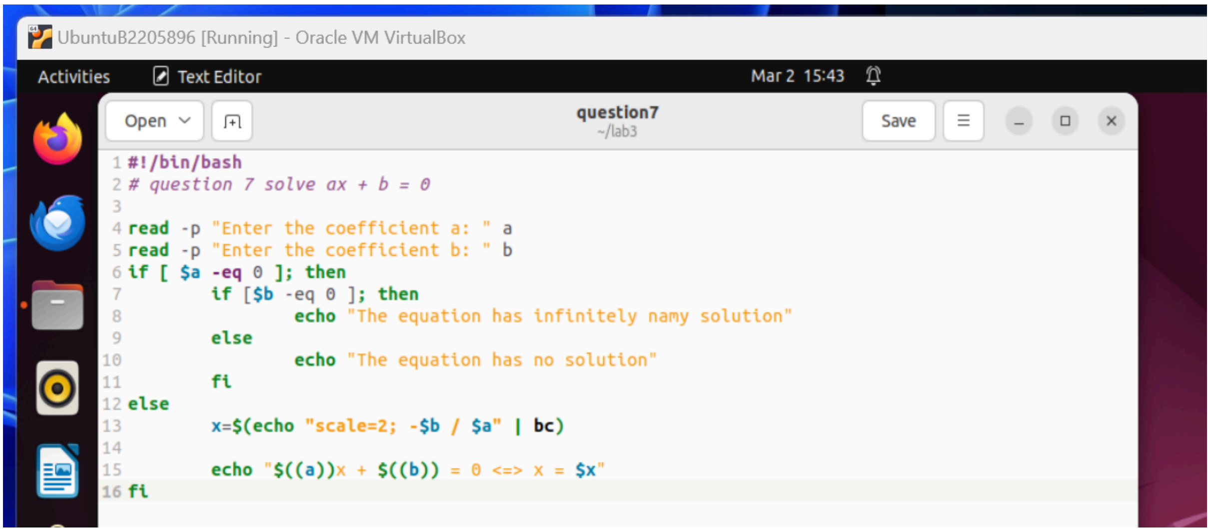


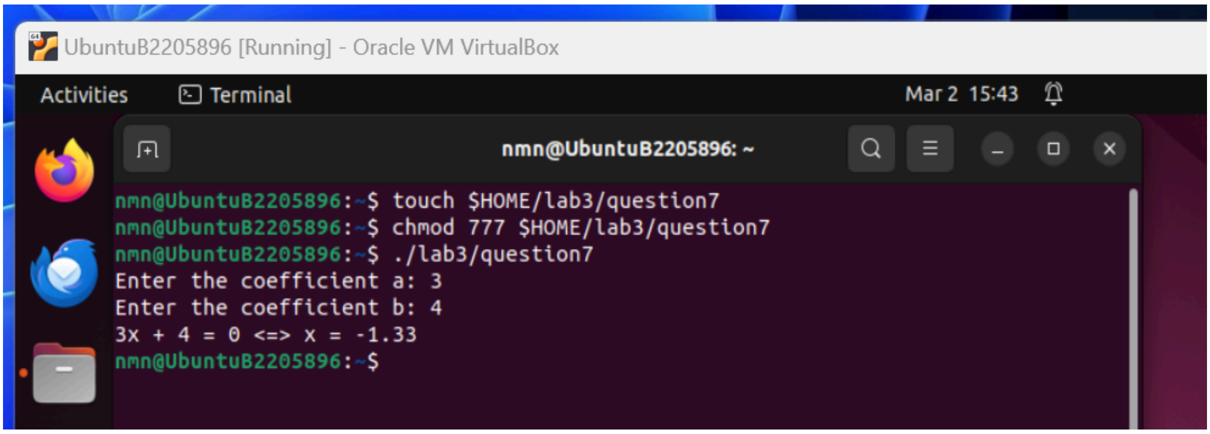


Question 7: Write a shell script to solve the equation *ax+b=0* where the coefficients *a* and *b* entered from the keyboard

Answer:

```
#!/bin/bash

# question 7 solve ax + b = 0

read -p "Enter the coefficient a: a

read -p "Enter the coefficient b: b

if [ $a -eq 0 ]; then

if [$b -eq 0 ]; then

echo "The equation has infinitely many solution"

else

echo "The equation has no solution"

fi

else

x=$(echo "scale=2; $b / $a" | bc)

echo "$((a))x + $((b)) = 0 <=> x = $x"

fi
```





Question 8: Write a shell script to print a multiplication table of a number entered by the user to the screen. After printing out the multiplication table, the program will ask the user to type "1" if the user wants to continue the process; otherwise, the process of printing the multiplication table will end.

Answer:

```
#!/bin/bash
#function to print multiplication table
show_table() {
num=$1
for i in [1..10]; do
echo "$num x $i = $((num*i))"
done
}
 # main
while true; do
read -p "Enter a number to print multiplication table: " number
show_table $number
```

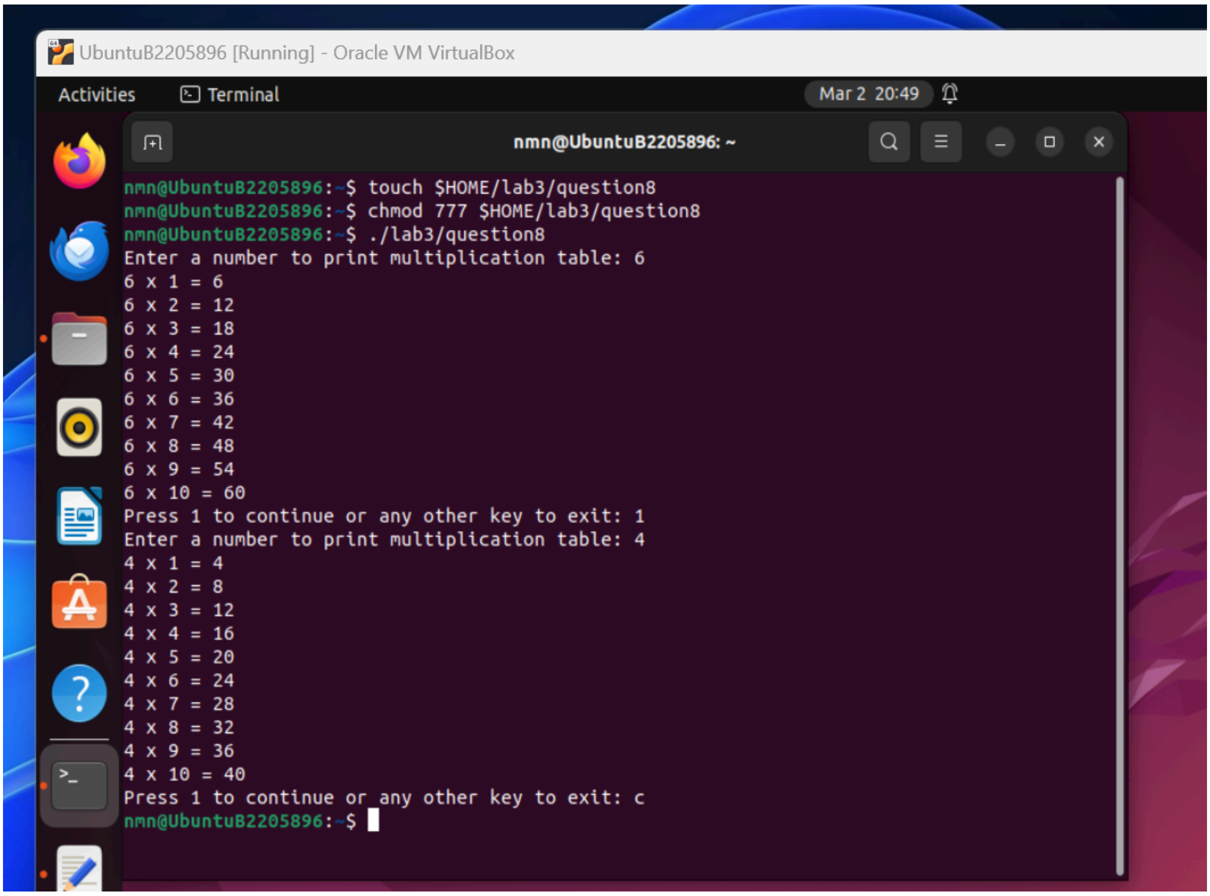
read -p "Press 1 to continue or any other key to exit: "key

if [[$key != 1]]; then

break

fi

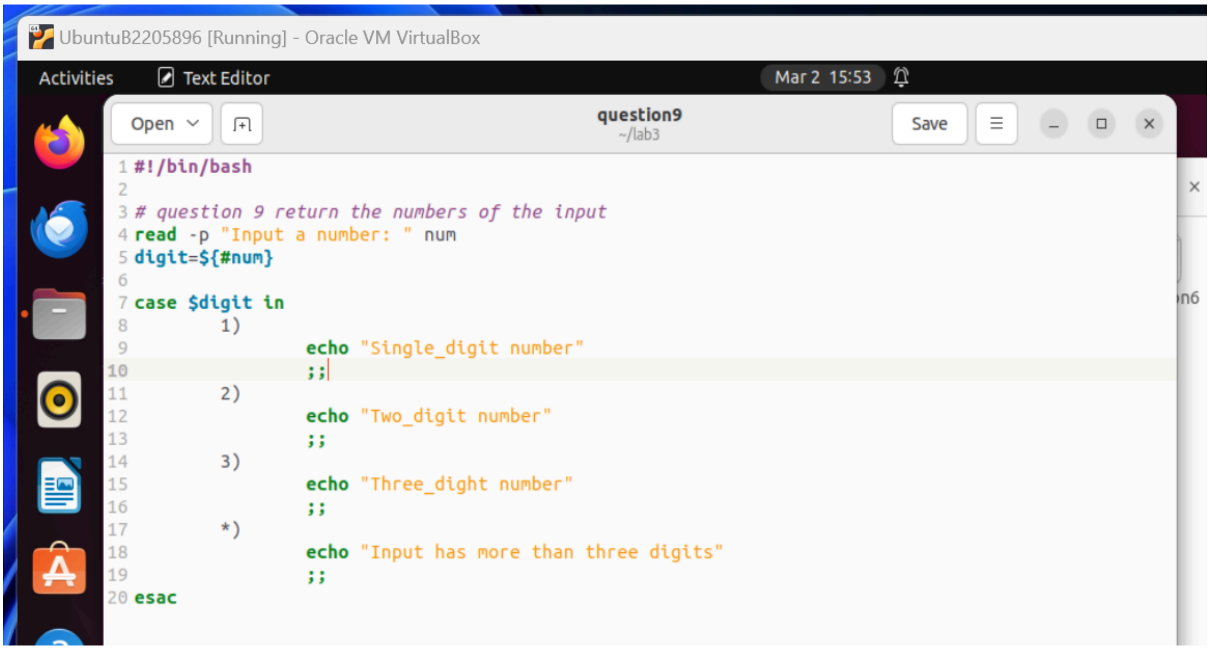
done

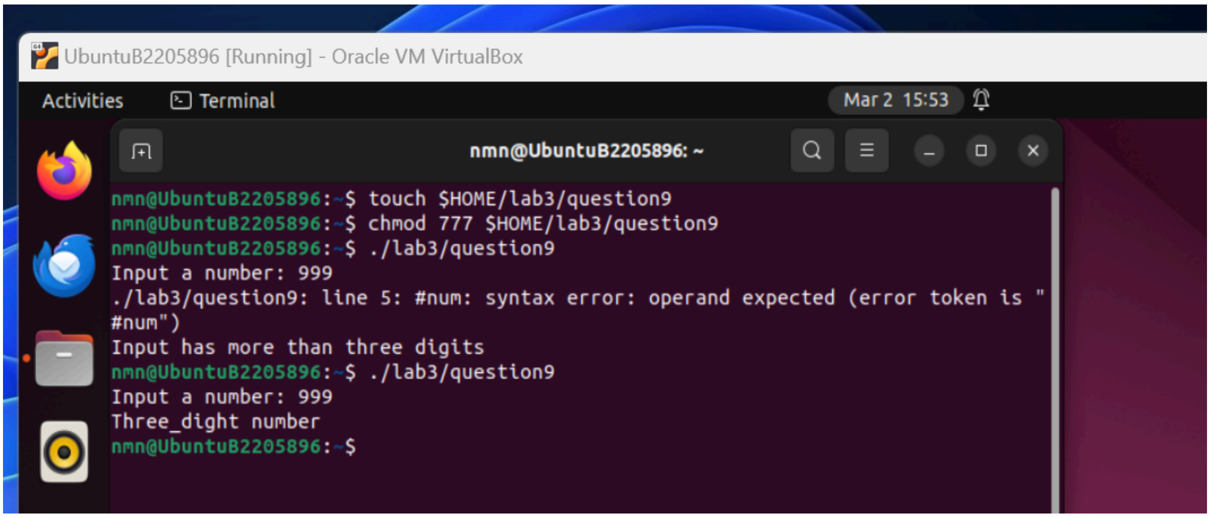


Question 9: Write a shell script that allows users to input an int number and returns the digit number of that number. For example, 10 → returns “two-digit number”; 100 → returns “three-digit number”.

Answer:

```
#!/bin/bash
# question 9 return the numbers of the input
read -p "Input a number: " num
digit=${#num}
case $digit in
1)
echo "Single_digit number"
;;
2)
echo "Two_digit number"
;;
3)
echo "Three_dight number"
;;
*)
    echo “Input has more than three digits”
    ;;
esac
```




UbuntuB2205896 [Running] - Oracle VM VirtualBox
Activities
Terminal
Mar 2 15:53
nmn@UbuntuB2205896: ~
nmn@UbuntuB2205896:~$ touch $HOME/lab3/question9
nmn@UbuntuB2205896:~$ chmod 777 $HOME/lab3/question9
nmn@UbuntuB2205896:~$ ./lab3/question9
Input a number: 999
./lab3/question9: line 5: #num: syntax error: operand expected (error token is "
#num")
Input has more than three digits
nmn@UbuntuB2205896:~$ ./lab3/question9
Input a number: 999
Three_dight number
nmn@UbuntuB2205896:~$

----------------------END----------------------